VeriCBAM: Quick User Guide for Companies & Institutions

30-Second Screening & Official PDF Audit Dossier Walkthrough

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Purpose: This guide provides verification bodies (DNV, TÜV, Bureau Veritas), customs authorities (DEHSt), and industrial importers with a zero-friction walkthrough to audit declared emissions in 30 seconds and export an official 1-page PDF Pre-Audit Screening Dossier.

1. Access the Quick Audit Cockpit

Open the live web application at https://vericbam-uqpz4jrnpteqetfvjxtslh.streamlit.app/. Ensure that Tab 1: Quick Audit & PDF Dossier Generator is selected in the navigation sidebar (default landing page).

2. Select Your Audit Testing Mode

- Option A (Pre-Loaded Reference Plant — Fastest): Select any of the 30 European cement or steel complexes (e.g., ThyssenKrupp Duisburg or Cementownia Górażdże). The system automatically pre-populates verified capacity, historical baselines, and multi-sensor satellite Earth observation signatures (Landsat 30m thermal ΔT, NASA VIIRS 375m active months, Planet VHR rail-yard contrast). Adjust declared production or emissions to test 'what-if' scenarios.
- Option B (External Foreign Supplier — Zero Prior History): Designed specifically for third-country exporters (Turkey, India, China, USA) lacking EU historical registries. Enter the plant name, sector (Cement / Steel), route (BF-BOF, Scrap-EAF, Dry Kiln), production volume, and declared CO_2. Check 'External Facility Lacks Past History (Use Satellite Data Only)' to evaluate the declaration purely via thermodynamic mass balance and satellite operational discrepancy.
- Option C (Upload CSV File): Drag and drop a standard declaration spreadsheet containing facility name, sector, route, production tonnes, and declared CO_2 for automated batch/single parsing.

3. Inspect the Automated Screening Diagnostic

The platform instantly computes and displays four diagnostic KPI cards:

- Declared Carbon Intensity: Declared t CO_2 / t product vs. the thermodynamic chemical lower bound (1.350 t/t for BF-BOF steel, 0.770 t/t for clinker).
- Physical Mass Balance: Displays FEASIBLE (green) or VIOLATION (red) with the exact missing CO_2 tonnage deficit.
- Satellite Kiln/Furnace Activity: Shows verified active operating months (out of 12) and localized core heat excess ΔT (°C).
- Inconsistency Risk Score: A calibrated probability score (0% to 100%) synthesized by the Bayesian log-odds engine:
  - CONSISTENT (< 25%): Green badge — Clear for expedited import filing without on-site inspection.
  - POTENTIAL INCONSISTENCY (25%–70%): Yellow badge — Request technical clarification, CEMS logs, and fuel receipts.
  - HIGH INCONSISTENCY RISK (> 70%): Red badge — Mandatory on-site audit recommended; freeze certificate clearance.

4. Export the Official Pre-Audit Screening Dossier (PDF)

At the bottom of Tab 1, click 'Download Official Pre-Audit Screening Dossier (PDF)'. The system instantly generates a publication-ready 1-page PDF document containing plant metadata, the legal verdict, the quantitative mass-balance table, multi-source evidence breakdown, and an official lead verifier SHA256 sign-off block—ready to attach to official customs records or audit working papers.